5. Data Inspection, condition, subsetting descriptive Statistic & Vi

Aim:-

To load the dataset, explore it, filter specific records an visualize score pattern using bar chart, histogram and boxplo

Algorithm:-

- Import pandas, matplotlib, seaborn
- Load the data into the dataframe.
- Explore structure using head, tail, shape, column, dtype, info d
- Filter students with math score > 70.
- Filter female students with math score > 70, select only gender & math score columns.
- Filter female students with master's degree and view descriptive statistics.
- Plot bar chart, histogram and boxplot with mean/median/mo

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from statistics import mode
df = pd.read_csv("Student performance.xls")
print [df.head(),\n df.tail()\n, df.shape(),\n df.columns.tolist()
        df.dtypes]
df.info()
print(df.describe())
math_above_70 = df.(df['math score'] > 70)
female_math_above_70 = df[(df['gender']=='female') &
                (df["math score"]>70)]
```

```
gender_math = df[["gender", "math score"]]
female_masters = df[(df["gender"] == "female") & (df["parental
    level of education"] == "master's degree")]
print(female_masters.describe())
df[["math score", "reading score", "writing score"]].mean().plot(
    kind="bar", color="skyblue"])
plt.title("Avg Score Per Subject");
plt.ylabel("Avg Score");
plt.show()
plt.hist(df["math score"], bins=10, color="orange", edgecolor="black")
plt.title("Distribution of Math Score");
plt.xlabel("Math Score");
plt.ylabel("Frequency");
plt.show()
sns.boxplot(x=df["math_score"], color="lightgreen")
plt.title("Spread of Math Scores");
plt.show()
mean_val, median_val, mode_val = df["math score"].mean(), df["math score"]
    .median(), mode(df["math score"])
plt.hist(df["math score"], bins=10, color="skyblue", edgecolor="black")
plt.axvline(mean_val, color="red", linestyle="--", label=f"Mean={mean_val:.2f}")
plt.axvline(median_val, color="green", linestyle="--", label=f"Median={median_val:.2f}")
plt.axvline(mode_val, color="purple", linestyle="--", label=f"Mode={mode_val [illegible]}")
plt.title("Math Score Histogram with Mean Median Mode");
plt.legend()
plt.show()
```

[illegible]

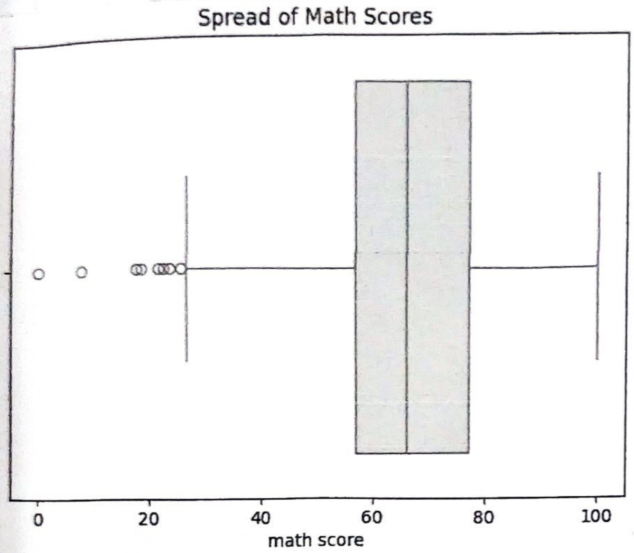
Output:

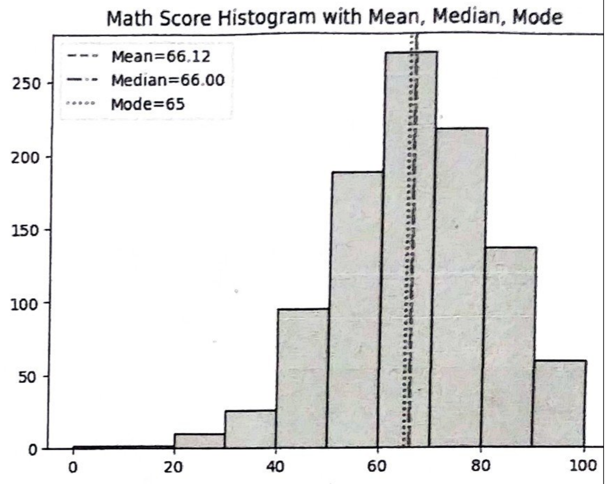
```
     Gender   race     Parent education.   lunch      Test courses   Math score   Reading   W
0    F        group B  Bachelor's Degree   Standard   none           72           72        7
1    F        group C  Some College        Standard   Completed      69           90        8
2    F        group A  Master's            Standard   none.          90           95        9

     Gender   race     Parent education.   lunch      Test course    Maths        Reading   W
1000 M        group D  Some Clg            Standard   none           76           64        66
1001 M        group C  Associate Degree    Standard   none           48           43        42
1002 F        group B  Bachelor's Degree   Standard   none           67           86        83
```

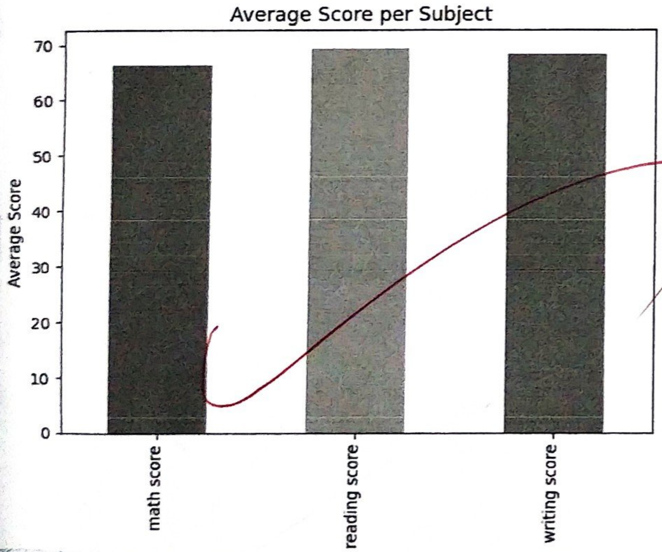
```
['gender', 'race/ethnicity', 'Parent level of education', 'lunch', 'writing Score
'Test preparation course', 'math Score', 'reading score',
```

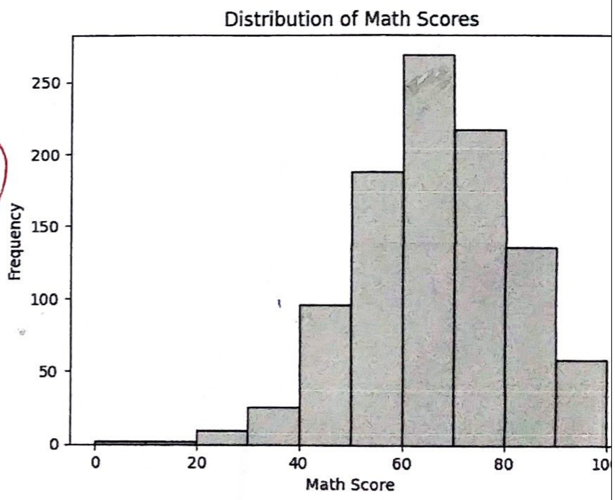
```
gender                object        Column              Non-Null    Dtyp
race                  Object        Gender              non-null    obje
Parent education      object        race                non-null    objec
lunch                 object        Parent education    non-null    objec
Test Course           object        lunch               non-null    obje
Math Course           object        Test Course         non-null    obje
reading Score         object        Math Score          non-null    obje
Writing Score         object        Reading Score       non-null    obje
                                    Writing Score       non-null    obj
dtypes: int64(3), objects(5)
```

```
         Math Score        Reading Score     Writing Sc
count    1005.000000       1005.000000       1005.00000
Mean     66.122388         69.183075         68.06667
Std      15.173234         14.614215         15.199095
min      0.000000          17.000000         10.000000
25%      57.000000         59.000000         58.0000
50%      66.000000         70.000000         69.0000
75%      77.000000         80.000000         79.0000
max      100.000000        100.000000        100.000
```









Result :-

The dataset was explored, filtered as required and visualiz

successfully.